

Retake Exam

1. Compute the following: **2p**

(a) $\lim_{n \rightarrow \infty} \frac{1^p + 2^p + \cdots + n^p}{n^{p+1}}, p > 0.$ (b) $\sum_{n \geq 1} \frac{n+1}{2^n}.$

2. Find the radius of convergence and the convergence set for the power series **2p**

$$\sum_{n \geq 2} \frac{x^n}{n \ln n}.$$

3. Let $f : \mathbb{R}^n \rightarrow \mathbb{R}$ be twice differentiable, with continuous second order partial derivatives, and let $x_0 \in \mathbb{R}^n$. Prove that for any $x \in \mathbb{R}^n$ there exists $c_x \in \mathbb{R}^n$ such that

$$f(x) = f(x_0) + \nabla f(x_0) \cdot (x - x_0) + \frac{1}{2}(x - x_0)^T H(c_x)(x - x_0).$$

Hint: Use Taylor's expansion for $g : [0, 1] \rightarrow \mathbb{R}$, $g(t) = f(x_0 + t(x - x_0))$. **1p**

4. Find and classify the stationary points over $\mathbb{R}^2$ of **2p**

$$f(x_1, x_2) = 2x_1^3 + 3x_2^2 + 3x_1^2x_2 - 24x_2.$$

5. Let $f : \mathbb{R}^3 \rightarrow \mathbb{R}$, $f(x) = x^T A x + 2b^T x$, with $A = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix}$, $b = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$, $x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$.

(a) Give a direction $d \in \mathbb{R}^3$ in which f decreases (descent direction) at $(1, 0, 1)$. **1p**

(b) Let $x \in \mathbb{R}^3$ and let $d \in \mathbb{R}^3$ be a descent direction of f at x . Find $t^* \geq 0$ for which **1p**

$$f(x + t^*d) = \min_{t \geq 0} f(x + td).$$

6. Find $\int_0^\infty \int_0^\infty e^{-(x^2+y^2)/2} dx dy$ using polar coordinates, then find $\int_0^\infty e^{-x^2/2} dx$. **1p**

Time: 2h. Total number of points: 10p. Please write your full name and your group.